

Non-parametric Tests

Statistical Methods Dossier with Subject-Specific Examples

1. Method Description

Nonparametric (distribution-free) tests do not assume a specific population distribution.

They work with ranks rather than raw values, making them robust to outliers and non-normality.

Trade-off: Lower statistical power when parametric assumptions are met.

Most common: Mann-Whitney U (2 independent groups), Wilcoxon signed-rank (2 paired groups), Kruskal-Wallis (3+ independent groups), Friedman (3+ paired groups).

2. Subject-Specific Examples

Social Sciences: Do two ethnic groups differ in social status ranking? Mann-Whitney U test. DV: ranked social status.

Economics: Do three investment strategies differ in median return? Kruskal-Wallis test.

Human Sciences: Do patients pain levels differ across 3 time points (pre, post, follow-up)? Friedman test.

Humanities: Do two authors differ in vocabulary richness (highly skewed)? Mann-Whitney U.

3. Important Concepts and Assumptions

Independent observations within and between groups.

Ordinal DV or continuous DV with severely violated normality.

Mann-Whitney: Tests whether one group tends to have larger values (not necessarily median difference).

Ties: Multiple identical values reduce test power; corrections exist.

4. Interpretation and practice

Report: 'A Mann-Whitney U test indicated that scores in Group A (median = 24.5) were significantly higher than in Group B (median = 18.2), $U = 234$, $z = -2.45$, $p = .014$, $r = .31$.'

Effect size: $r = Z / \sqrt{N}$. Benchmarks: 0.1 (small), 0.3 (medium), 0.5 (large).

Kruskal-Wallis is the nonparametric equivalent of one-way ANOVA; Friedman of repeated-measures ANOVA.

5. Code Examples

R: `wilcox.test(y ~ group, data = d)` # Mann-Whitney; `wilcox.test(pre, post, paired = TRUE)`; `kruskal.test(y ~ group, data = d)`;
`friedman.test(y ~ time | id, data = d)`

Python: `from scipy import stats; stats.mannwhitneyu(g1, g2); stats.wilcoxon(pre, post); stats.kruskal(g1, g2, g3);`
`stats.friedmanchisquare(t1, t2, t3)`

SPSS: NPAR TESTS /M-W = y BY group(0 1). NPAR TESTS /WILCOXON = pre WITH post.